

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Spring compression tester, Part Number 5299714.

Additional Service Items

- No additional service items required.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

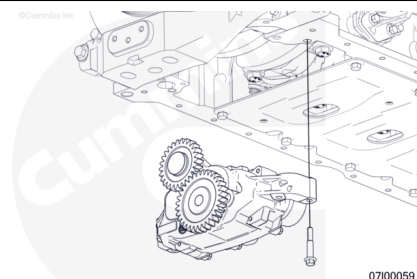
WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-037.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-025.html)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-035.html)

Remove

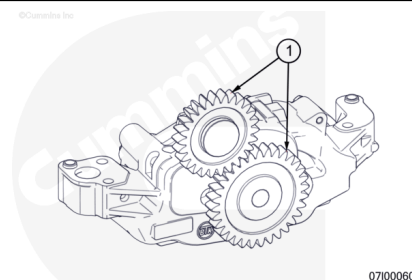
Remove the mounting capscrews and oil pump.



Disassemble

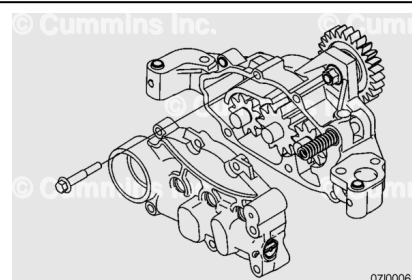
⚠ WARNING ⚠

Wear appropriate eye and face protection when performing this procedure. Springs are under tension and can act as a projectile if released, causing personal injury.

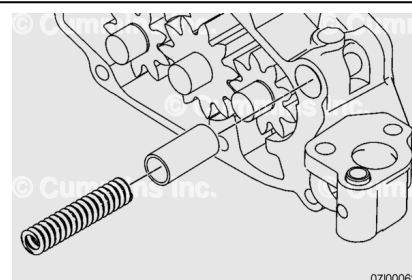


Note : The outside idler and drive gears (1) are press fitted to the pump housing. These gears can not be removed from the pump housing.

Remove the assembly capscrews and the lubricating oil pump cover.

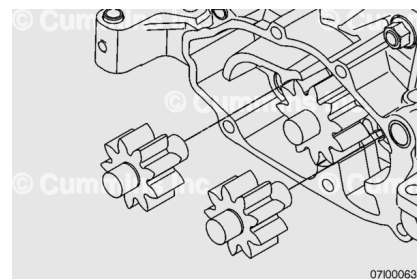


Remove the high-pressure relief spring and plunger.

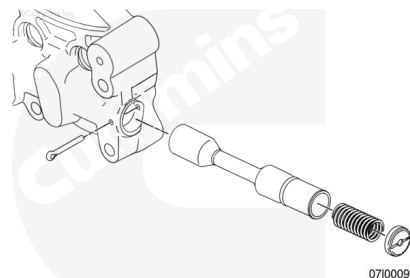


Remove the internal idler pumping gear and shaft assemblies from the front lubricating oil pump body.

Note : The external and internal drive gears are permanently assembled together through the center shaft. These parts are not designed to be disassembled from the front pump housing.



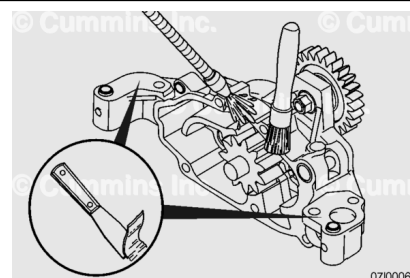
Remove the lubricating oil pressure regulator and spring by removing the split pin that secures the spring cap to the lubricating oil pump.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

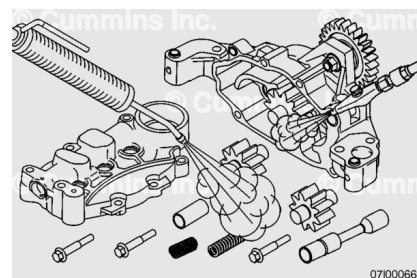
Do not allow dirt or material to enter the oil passages.
Bearing damage can result.

Clean the lubricating oil pump components. Use solvent.

Use a bristle brush to clean any internal ports, internal drive gears, and shafts.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

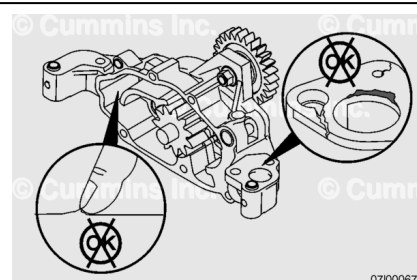
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a steam cleaner to rinse the components.

Dry with compressed air.

Inspect the following areas of the lubricating oil pump for damage or blockage:

- Lubricating oil pump body gear cavities
- Inlet and outlet ports
- Internal oil passages
- Front and back lubricating oil pump housing bodies
- Pressure regulator bore
- High-pressure relief bore and seat.

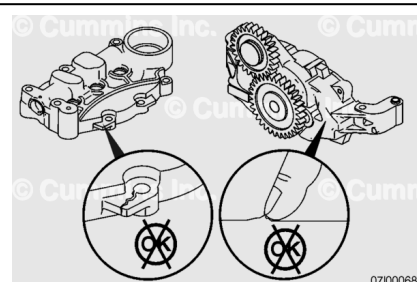


If the lubricating pump housings are damaged, the pump **must** be replaced.

Inspect the front and rear lubricating oil pump covers for damage.

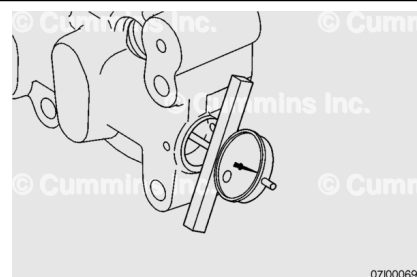
Inspect the bushings for scuffing and wear.

If marks on the lubricating pump covers can be felt with a fingernail, the part **must** be replaced.



Measure the regulator spring bore for beat-in with the regulator plunger in the bore.

Spring Bore Depth Specification		
mm		in
38.5	MAX	1.52
36.5	MIN	1.44



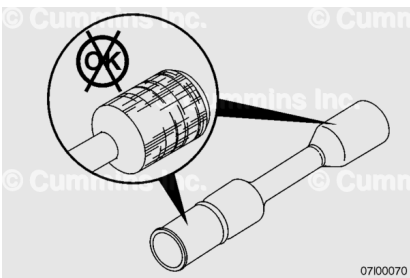
If the regulator spring bore depth is **not** within the specifications, the housing **must** be replaced.

Inspect the pressure regulator spool for scuffing.

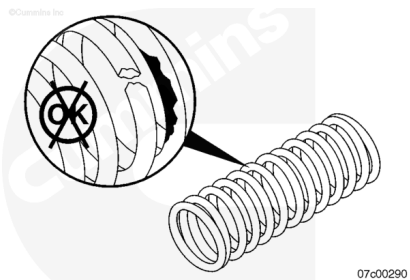
Minor spool scuff is acceptable. If the scuff can be felt by a fingernail, the spool will need to be replaced.

Insert the spool in the bore and check for smooth movement in and out of the bore. If any hesitation is felt, the spool **must** be replaced and the bore inspected for burrs.

Remove any burrs found in the bore before installing a new spool.

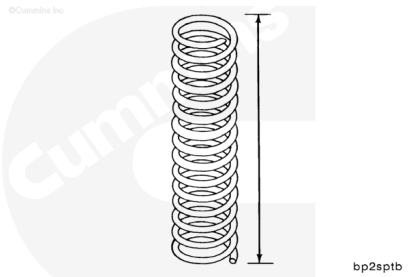


Inspect the pressure regulator spring for a broken coil or rust.



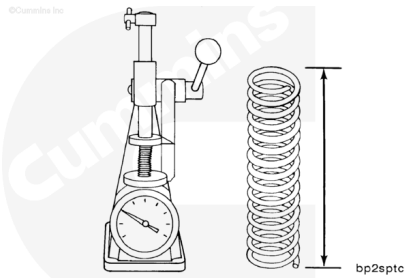
Measure the free length of the pressure regulator spring.

Oil Pressure Regulator Spring Free Length		
mm		in
42.8	NOM	1.69



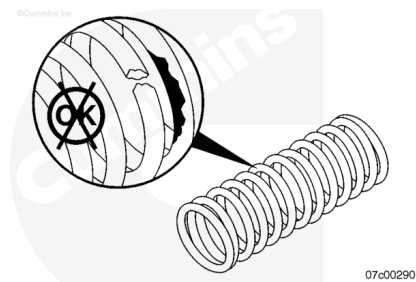
Use spring compression tester, Part Number 5299714, to measure the pressure regulator valve spring at working height, 33.8 mm [1.33 in].

Lubricating Oil Pressure Regulator Spring Load at 33.8 mm [1.33 in]		
n.m		ft-lb
84.55	MIN	19.01
93.45	MAX	21.01



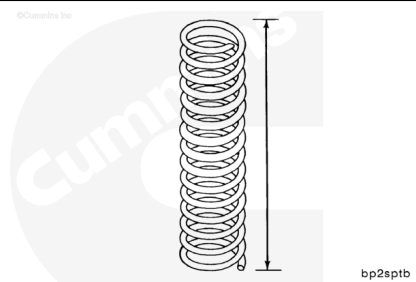
If the regulator spring is **not** within the specifications, the spring **must** be replaced.

Inspect the high-pressure relief spring for a broken coil or rust.



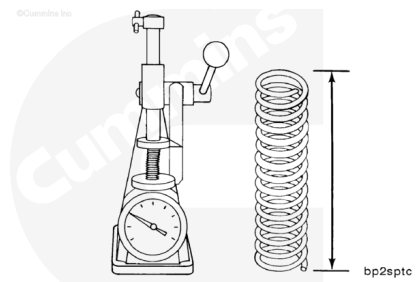
Measure the free length of the high-pressure relief spring.

High-Pressure Relief Valve Spring Free Length		
mm		in
87.2	NOM	3.43



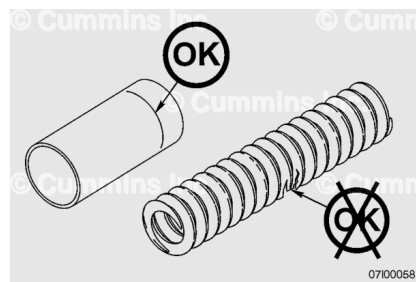
Use spring compression tester, Part Number 5299714, to measure the high-pressure relief spring at working height, 75.07 mm [2.956 in].

High-Pressure Relief Valve Spring (Compressed Length) Load at 75.07 mm [2.956 in]		
n		ft-lb
283.4	MIN	63.71
332.6	MAX	74.77



If the pressure relief valve spring is **not** within the specification, the spring **must** be replaced.

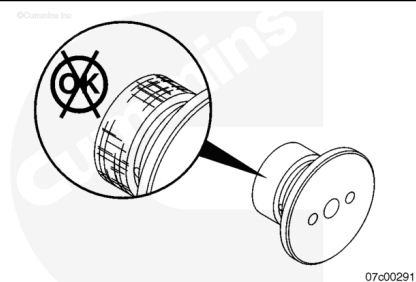
Inspect the high-pressure relief plunger. A slight wear around the plunger from contact with the pump body and spring is acceptable.



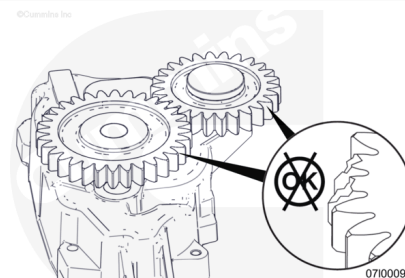
Discard the mounting bolt.

Inspect the external idler gear retainer.

Minor retainer scuffing is acceptable. If the scuffing can be felt by a fingernail, replace the retainer.

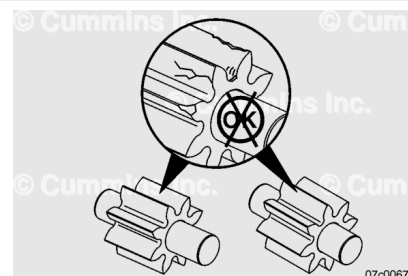


Inspect the external drive gears for cracks, chips, scuffing, and broken teeth.



Inspect the lubricating oil pump internal gears and their bushings for excessive wear, cracks, or broken teeth.

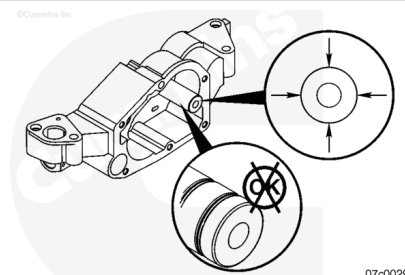
Erosion and pitting on the tip and root of the gear tooth is unacceptable. The gear **must** be replaced.



Inspect the internal driven gear and shaft for damage.

Measure the outside diameter of the idler shaft.

Internal Drive Gear Shaft Outside Diameter		
mm		in
19.960	MIN	0.7855
20.053	MAX	0.7891

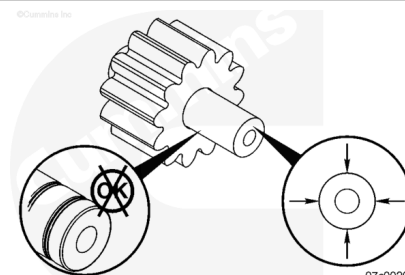


If the drive shaft is **not** within specifications, it **must** be replaced.

Inspect the internal drive gear and shaft assembly for damage.

Measure the outside diameter of the internal drive gear shaft.

Internal Driven Gear Shaft Outside Diameter		
mm		in
19.960	MIN	0.7855
20.053	MAX	0.7891

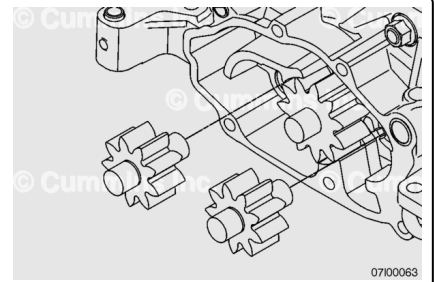


If the drive shaft is **not** within specifications, the gear assembly **must** be replaced.

Assemble

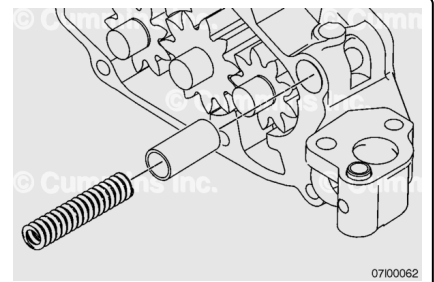
Lubricate the drive shaft. Use clean engine lubricating oil.

Install the internal drive gears onto the housing.



Install the plunger first, then the spring.

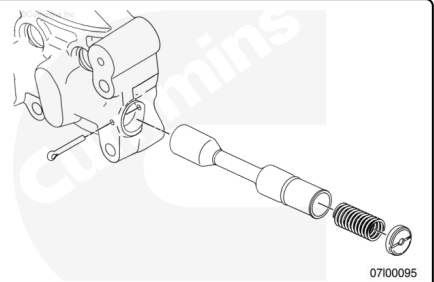
Install the high-pressure relief and spring into its bore in the lube pump housing.



Lubricate the oil pressure regulator valve and bore with clean engine oil.

Install the valve, spring, and cap.

Use a new split pin to secure the cap to the lubricating oil pump body.



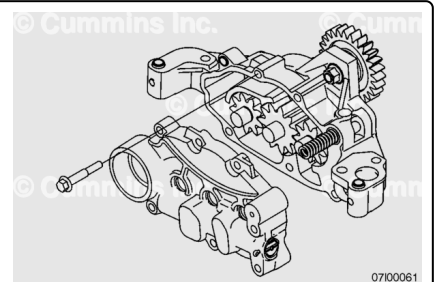
No gasket is used when installing the lube pump cover.

Install the cover to the lube pump.

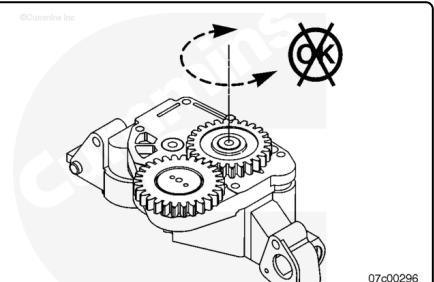
Tighten the center capscrews first, then the outside capscrews.

Install the capscrews and tighten.

Torque Value: 20 n•m [14.75 ft-lb]

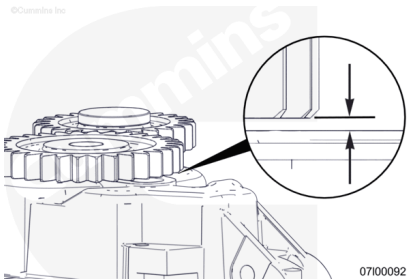


Turn the main drive gear by hand to check the gears for freedom of rotation.



Measure the clearance between the drive gear and pump housing.

Clearance		
mm		in
1.04	MIN	0.0252
2.36	MAX	0.077



Install

⚠ CAUTION ⚠

Improper gear lash can cause damage to the gear train and lubricating oil pump.

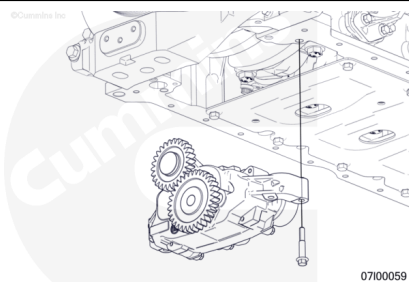
Align the dowels on the oil pump with the mounting holes in the cylinder block.

Install the oil pump and the capscrews.

Tighten the capscrews.

Torque Value: 56 n•m [41 ft-lb]

Note : The lubricating oil pump on the engine does not use shims. Measuring the gear backlash is not necessary.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-035.html>)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-025.html>)

- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/377/377-007-037.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for leaks.

Last Modified: 26-Sep-2022
